Lesson notes · for you

Meet the microcontroller

IoT · Level 2 · Lesson 4

Materials
Paper and a pen; A torch and a book, to act it out; Optional: a phone or computer with a browser, for MakeCode; Optional: a BBC micro:bit and a USB cable

Learning goals

- Say what a microcontroller is and where you find one.
- Explain a program that reads a sensor, decides and acts, again and again.
- Write a night-light program as blocks or a flowchart.

Activity: Night light, three ways

1. On paper: a box for “read the light level”, a diamond for “below 50?”, boxes for LED on and LED off, and an arrow back to the read box.
2. A friend shades a window or a torch with a book. You read the light, decide with your flowchart, and switch a torch on or off; then wait and repeat.
3. Open MakeCode for micro:bit in a browser. Put a light-level check inside the forever block, with if, then and else blocks.
4. In the simulator, drag the light slider down, then up. Does the LED (or the screen) switch at the right moment?
5. 50 is only an example. Try 20 and 100 and see when the light comes on. Got a micro:bit? Read your room at dusk and pick your own number.

Power the micro:bit from USB or its own battery pack only, never from mains. Ask before you use a family phone or computer.

Discussion

- Which devices around you run a loop like this?
- Why is it useful that the threshold is a number you can change?
- What could go wrong if the light sensor sat right next to the LED?

Check yourself

- What happens to a night-light program with no loop?

Try it this week
Night light, three ways: Draw your night light as a flowchart on paper: a box to read the light, a diamond for the question, boxes for LED on and LED off, and an arrow back to the start. Then build it in MakeCode, free in a browser, and test it with the simulator's light slider. Got a micro:bit? Show the light level on its screen, note the reading in your room at dusk, and use a number just above it as your threshold. No phone or computer? Be the microcontroller: a friend changes the light with a book, and you follow your flowchart with a torch.

Worksheet

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Name _____ Class _____ Date _____

1 **Tunde's night-light program has no forever loop: it reads the light once, when he switches it on in the afternoon. What happens at night?**

- ◯ The LED flickers all night, because it reads only now and then
- ◯ Nothing: it read once, in daylight, so the LED stays off
- ◯ The LED comes on when it gets dark, as a night light should

2 **Which of these have a microcontroller inside?**

- ☐ A TV remote
- ☐ A prepaid electricity meter
- ☐ A wind-up wall clock with no battery
- ☐ A washing machine with a program dial and a display
- ☐ A torch with just a switch, a battery and a bulb

3 **A microcontroller needs the internet to work.**

◯ True ◯ False

4 **Complete the sentence about reading a light sensor.**

A micro:bit gives the light level as one byte, from 0 to _____. An Arduino Uno uses 10 bits, from 0 to _____.

Word bank: 255 · 256 · 1024 · 1023

5 **Put the night light's flowchart in order, from the start.**

- ☐ Is it below 50?
- ☐ Wait a moment, then go back to read
- ☐ Read the light level
- ☐ Start: set up once
- ☐ Yes: LED on. No: LED off

✎ **Which device at home or in your town has a microcontroller inside? What does it read, decide and do?**

Check your answers

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1 **Nothing: it read once, in daylight, so the LED stays off**

Without the loop, the program read the light once, saw daylight, switched the LED off and stopped.

2 **A TV remote; A prepaid electricity meter; A washing machine with a program dial and a display**

The remote, the meter and the washing machine all read inputs, decide and act, so each needs a small computer.

3 **False**

Most microcontrollers never connect at all: a remote or a washing machine runs its program all alone.

4 **A micro:bit gives the light level as one byte, from 0 to 255. An Arduino Uno uses 10 bits, from 0 to 1023.**

One byte holds 256 values, and starting from 0 the biggest is 255.

5 **Top to bottom: 3, 5, 2, 1, 4**

Setup happens once. Then the loop reads first, decides on that fresh reading, acts, and waits before going round again.